

ML Model Features & Statutory LARR Act 2013 Legal Guide

Department of Land Resources (DoLR) · Ministry of Rural Development · Government of India

1. Exhaustive Breakdown of the 14 ML Model Features

The XGBoost model processes 14 engineered features representing physical, statutory, financial, and administrative friction points:

#	Feature Name	Type & Units	Statutory & Operational Meaning	Why Used & Impact on Delay Prediction
1	total_acreage_ha	Float (ha)	Total physical footprint required for corridor.	Larger footprints span multiple taluks, increasing joint measurement survey complexity.
2	num_land_parcels	Integer	Cadastral fragmentation (khasra/survey plots).	High fragmentation multiplies title verification latency and citizen objections.
3	private_to_govt_ratio	Float (Ratio)	Ratio of private titled land to government land.	Govt land transfers quickly; private land requires formal notice, SIA, and consent.
4	sc_st_land_percentage	Float (%)	Tribal protective safeguards (Sec 41/42).	Requires special Tribal Development Plan, 1/3 upfront compensation, and Gram Sabha consent.
5	multi_crop_irrigated_percentage	Float (%)	Food security ceiling on farmland (Sec 10).	Statutory limits on multi-crop land; triggers High Court stay orders if ceilings are breached.
6	required_consent_percentage	Integer (%)	Prior consent hurdle (0%, 70%, 80%).	Mandatory 70% for PPP or 80% for Private; failure to achieve threshold halts project.
7	non_owner_to_owner_paf_ratio	Float (Ratio)	Livelihood claim density (Sec 3(c)).	Non-owners (tenants, labourers) are entitled to R&R; high density sparks census disputes.
8	circle_rate_disparity_ratio	Float (Ratio)	Market value vs Circle rate ratio.	Single largest delay driver; if disparity >1.5x, landowners reject award and seek court stay.
9	rr_cost_share_percentage	Float (%)	R&R; budget share of total project cost.	Higher R&R; overheads require multi-agency coordination (housing boards, DoLR).
10	rural_multiplier_factor	Float (1.0 - 2.0)	Rural market value multiplier (Sec 26(2)).	Ambiguity in state distance notifications triggers litigation over calculated base compensation.
11	district_litigation_rate	Float (%)	District historical land stay rate (5-yr).	Captures localized litigiousness, presence of active farmer unions, and local stay precedents.
12	revenue_staff_vacancy_rate	Float (%)	Survey/revenue staff vacancy rate.	Directly bottlenecks Section 12 field demarcation and Section 15 objection disposal.
13	avg_s15_resolution_days	Float (days)	Section 15 objection hearing latency.	Slow disposal directly eats into the strict 12-month statutory window before Sec 19.
14	days_since_s11	Integer (days)	Days elapsed since Section 11 preliminary notice.	Statutory 12-Month Clock: If days > 365 without Sec 19 declaration, acquisition lapses by law.

2. LARR Act 2013 Statutory Legal Framework & Key Mandates

The Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act, 2013 governs all acquisitions in India. Below are the mandatory statutory checkpoints:

Section / Provision	Statutory Rule & Threshold	Mandatory Compliance & Consequences of Violation
Section 4 & 7 (SIA Study)	Mandatory Social Impact Assessment in consultation with Gram Sabha.	Independent Multi-Disciplinary Expert Group must approve SIA within 6 months. Non-compliance voids Section 11 notice.
Section 10 (Food Security)	Multi-crop irrigated land acquisition permitted only as demonstrable last resort.	Must not exceed cumulative district ceiling. Failure to prove lack of alternative barren land triggers High Court stays.
Section 11 (Preliminary Notice)	Official Gazette publication declaring intent to acquire.	Freezes all land transactions/sales. Starts the strict 12-month statutory clock for Section 19(2).

Section 15 (Hearing Objections)	60-day window for citizens to file written objections.	Collector must give personal hearing to objectors. Denial of natural justice leads to writ petitions under Article 226.
Section 19(2) (12-Month Lapsing)	MANDATORY RULE: Final declaration must be published within 12 months (365 days) of Section 11 notice.	AUTOMATIC LAPSE: If 365 days elapse without Section 19 publication, preliminary notification lapses and acquisition dies.
Section 26 & Proviso 4 (Market Value)	Base market value determined via highest 50% sale deeds or circle rate.	Proviso 4 empowers Collector to convene direct negotiation committees when circle rates are visibly outdated.
Section 30 (100% Solatium)	Mandatory 100% Solatium added over total calculated compensation.	Non-negotiable statutory multiplier (Total Award = Base × Multiplier + 100% Solatium).
Section 38 (Possession)	Physical possession prohibited until full compensation and R&R; are disbursed.	Attempting possession before monetary disbursement is illegal and attracts contempt of court.
Sections 41 & 42 (SC/ST Tribal Land)	Prohibits acquisition in Scheduled (Fifth Schedule) areas as far as possible.	Requires Gram Sabha prior consent, 1/3 upfront compensation, and dedicated Tribal Development Plan.
Section 64 (Court References)	Aggrieved owners can seek reference to LARR Authority (District Judge).	Does not stay work unless an injunction is granted; causes massive delayed compensation enhancement liabilities.